Hugo Marquerie 14/06/2025

Esperanza condicionada a σ-álgebras anidadas

Proposición 1. *Sea $X \in \mathcal{L}^1(\mathbb{P})$ y $\mathcal{F}_1, \mathcal{F}_2$ σ-álgebras tales que $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \Sigma$*

$$\implies \mathbb{E}\left[\mathbb{E}\left[X \mid \mathcal{F}_1\right] \mid \mathcal{F}_2\right] = \mathbb{E}\left[\mathbb{E}\left[X \mid \mathcal{F}_2\right] \mid \mathcal{F}_1\right] = \mathbb{E}\left[X \mid \mathcal{F}_1\right] \text{ c.s.}$$

***Demostración*:** Como $\mathcal{F}_1 \subset \mathcal{F}_2$, entonces $\mathbb{E}[X \mid \mathcal{F}_1]$ es $\mathcal{F}_2$-medible. Por lo tanto, por la propiedad `lem-esperanza-condicionada/`**??** del `lem-esperanza-condicionada/`**??**, tenemos que

$$\mathbb{E}\left[\mathbb{E}\left[X \mid \mathcal{F}_1\right] \mid \mathcal{F}_2\right] = \mathbb{E}\left[X \mid \mathcal{F}_1\right] \text{ c.s.}$$

Para la otra igualdad, sea Z $\mathcal{F}_1$-medible, entonces también es $\mathcal{F}_2$-medible.

$$\implies \mathbb{E}\left[Z \cdot \mathbb{E}\left[X \mid \mathcal{F}_1\right]\right] \overset{\substack{Z \text{ es } \mathcal{F}_1\text{-medible} \\ \downarrow}}{=} \mathbb{E}\left[X \cdot Z\right] \underset{\substack{\uparrow \\ Z \text{ es } \mathcal{F}_2\text{-medible}}}{=} \mathbb{E}\Big[Z \cdot \overbrace{\mathbb{E}\left[X \mid \mathcal{F}_2\right]}^{\mathcal{F}_2\text{-medible}}\Big] \overset{\substack{Z \text{ es } \mathcal{F}_1\text{-medible} \\ \downarrow}}{=} \mathbb{E}\Big[Z \cdot \mathbb{E}\big[\mathbb{E}\left[X \mid \mathcal{F}_2\right] \mid \mathcal{F}_1\big]\Big]$$

Por tanto, por unicidad de la esperanza condicionada, queda demostrado. ∎

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